

22 · The reader-training cost difference was large

Whole-training process time for the same 300 updates, including construction and caching. Each pair starts from matched parameters.

Training seed	Backpropagation	Error-inference training
Seed 0	0.241 h	24.579 h
Seed 1	0.252 h	24.112 h
Seed 2	0.242 h	23.432 h

Measured paired cost ratios: 102.1×, 95.8× and 96.7×. Backpropagation took about 14–15 minutes per reader; ePC took about a day.

The installed ePC training procedure was substantially more expensive, without a consistent efficacy-and-harm gain.

Source: Completed PC-reader process receipts. Acquisition and evaluation are separate; bars share a zero origin and scale.